Strategic Fee Choice and Transaction-Priority Allocation under Incomplete Information

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Nonbinding recommendations improve priority alignment (87.11% → 97.99%) but increase fees (2.352 → 2.812) and reduce user welfare (3.057 → 2.678). Effective coordination requires balancing allocation accuracy, costs, and information reliability.

01 Question and foundations

Research question & verified gap

Research question: How do private urgency, accumulated waiting, and nonbinding fee recommendations affect transaction-priority allocation, fee expenditure, and user welfare?

Background & Application: Blockchain transaction processing involves scarce confirmation capacity, fee competition, and heterogeneous waiting costs.

Verified Gap: Existing studies examine fee competition (Huberman et al., 2021), dynamic posted prices (Ferreira et al., 2021), and payment redesign (Lavi et al., 2022). FP7 instead evaluates differentiated, nonbinding recommendations under fixed allocation and payment rules.

Strategic model and Intellectual Merits

1. Strategic model

Players: Two users with private Urgent or Patient types; Pr(Urgent) = 0.6.

Actions: Low (1), Medium (2), or High (3) fees over three connected rounds.

Allocation & Payment: The higher fee receives confirmation priority; ties are broken randomly. The confirmed user pays the selected fee; unconfirmed users accumulate waiting.

Comparison: Complete-information coordination, incomplete-information policies, and nonbinding recommendations (ρ = 0.85), with a 3% action tremble.

2. Intellectual Merits & Evidence Boundary

Contribution: A controlled comparison of information conditions and nonbinding recommendations under unchanged allocation and payment rules.

Evidence Boundary: Results come from simulated decision policies, not observed participant behavior or a demonstrated dynamic equilibrium.

Three complementary lenses

Game Theory: Strategic fee choices under private urgency and a stage-game Nash benchmark.

Social Choice: Priority alignment, waiting costs, fee expenditure, and user welfare.

Mechanism Design: Differentiated, nonbinding recommendations without changing allocation or payment rules.

Experimental Comparison

Complete Information (CB): Coordinated fee choices using users' urgency and waiting states.

Incomplete Information (INC): Independent fee choices based on each user's private information.

Nonbinding Signal (SIG): Differentiated recommendations followed with probability ρ = 0.85.

02 Model and verified evidence

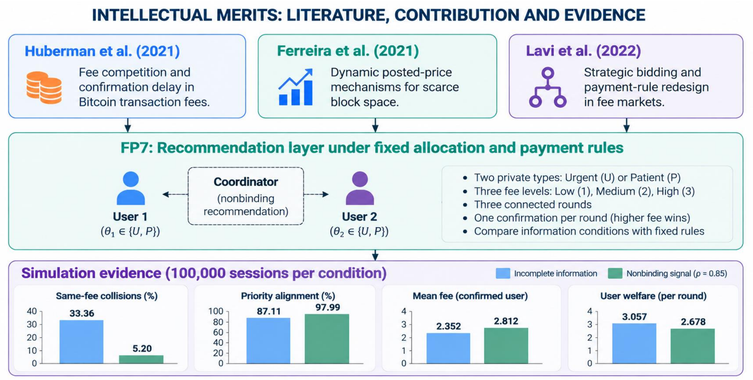


Figure 1. Building on fee competition (Huberman et al., 2021), dynamic pricing (Ferreira et al., 2021), and payment redesign (Lavi et al., 2022), FP7 evaluates nonbinding recommendations under fixed rules, supported by simulation evidence.

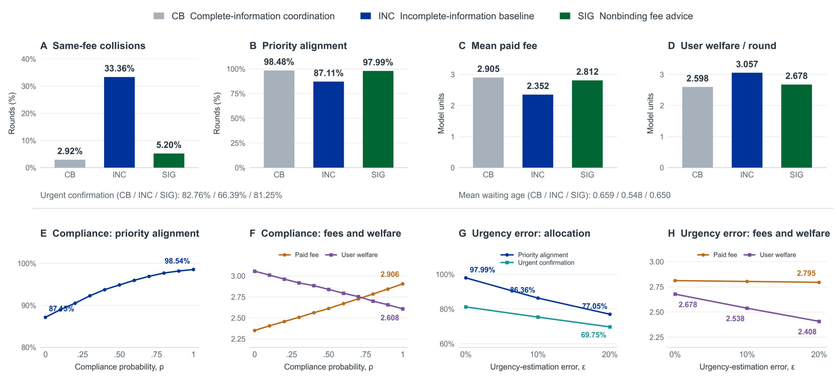


Figure 2. Simulation results across three information conditions, recommendation-compliance probabilities, and urgency-estimation errors. Nonbinding advice improves priority alignment but increases fees and reduces combined user welfare.

Interpretation: Better coordination does not guarantee higher welfare. Reduced fee collisions can coexist with higher payments and waiting costs. Welfare is the sum of both users' payoffs per round, excluding the fee recipient.

Verification: Google Colab simulation; 100,000 sessions per condition; GitHub commit 1793e2a

Evidence status: Observed computational results under specified simulation policies; behavioral validation remains planned.

03 Practical impacts and roadmap

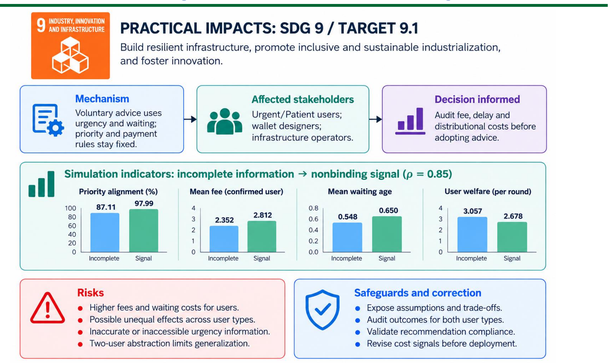


Figure 3. Practical implications for SDG 9, Target 9.1: evaluating priority alignment, user costs, and welfare under nonbinding fee recommendations.

Key Finding: Priority alignment improves from 87.11% to 97.99%, but mean fees increase from 2.352 to 2.812 and user welfare declines from 3.057 to 2.678.

Limitation: These are simulation results, not measured real-world SDG impacts. Deployment requires cost transparency, fairness audits, and behavioral validation.

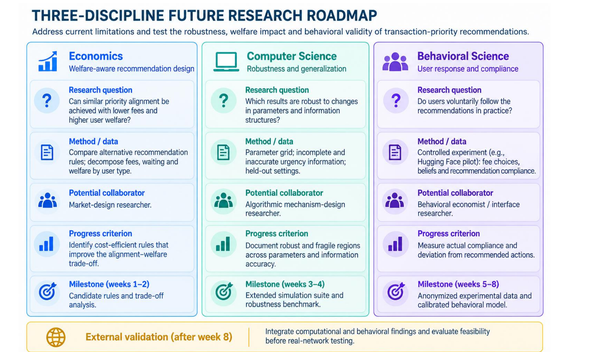


Figure 4. Future research across economics, computer science, and behavioral science, followed by external validation.

Next Computational Test: Following the suggestion, compare fixed-probability compliance with an incentive-based policy in which users follow recommendations only when expected priority benefits justify additional fees.

Integration and updated named contributions

Research Integration: Combine economic welfare criteria, computational robustness, and observed behavioral responses to develop cost-efficient priority recommendations.

Lin Zhang evaluated allocation, fees, welfare, compliance sensitivity, and urgency-estimation errors; revised and verified Colab outputs; developed the initial Hugging Face interface and A0 poster.

Zijie Wang developed the formal fee-choice model, three-round simulation, GitHub repository, and updated Hugging Face game; verified model implementation.

Limitations: Synthetic decision policies, assumed compliance, and a two-user setting limit external validity. Future work will test incentive-based compliance and collect participant-level evidence.

Acknowledgments

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Selected references

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[2] Ferreira, M.V.X., Moroz, D.J., Parkes, D.C. & Stern, M. (2021). Dynamic Posted-Price Mechanisms for the Blockchain Transaction-Fee Market. AFT, 86–99. doi:10.1145/3479722.3480991.

[3] Lavi, R., Sattath, O. & Zohar, A. (2022). Redesigning Bitcoin's Fee Market. ACM TEAC 10(1), Art.5. doi:10.1145/3530799.

[4] United Nations. Goal 9: Industry, Innovation and Infrastructure, Target 9.1. https://sdgs.un.org/goals/goal9

[5] United Nations. Sustainable Development Goals: Communications Materials. https://www.un.org/sustainabledevelopment/news/communications-material/

OPEN MATERIALS

Google Colab: https://colab.research.google.com/drive/1KANyhcqVYuxHoduXRjxSqYtpzV17nu_w?usp=sharing

Hugging Face: huggingface.co/spaces/dku-comsci-econ206-2026/SubmitOrWait

GitHub: github.com/dku-comsci-econ206-Autumn2026/

PS2-FP7-ZijieWang-LinZhang (GitHub commit 1793e2a)

Hugging face game

Github Repo